Review of Key Formulas for Defining Meal Boundaries in NR Phase of Order 1 Mice

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October 29, 2024

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1 Introduction

This document provides a brief review of the key mathematical formulas employed in defining meal boundaries in the NR (No-Restriction) phase of Order 1 mice based on pellet consumption data. Each formula is explained in terms of its purpose and role within the overall analysis framework.

2 Key Formulas

2.1 1. Inter-Pellet Interval (IPI) Calculation

$$\text{IPI}_i = \text{Timestamp}_{i+1} - \text{Timestamp}_i \tag{1}$$

Purpose: Calculates the time difference between consecutive pellet consumption events. This interval helps in determining whether pellets belong to the same feeding cluster (meal) or mark the boundary between separate meals.

Role in Analysis: Establishes the foundation for clustering pellets based on temporal proximity, which is crucial for accurate meal segmentation.

2.2 2. Kernel Density Estimation (KDE)

$$\hat{f}(x) = \frac{1}{nh} \sum_{i=1}^{n} K\left(\frac{x - x_i}{h}\right) \tag{2}$$

Purpose: Estimates the probability density function (PDF) of the IPI data in a non-parametric way, smoothing out the distribution to identify underlying patterns.

Role in Analysis: Helps visualize the distribution of IPIs to identify natural cutoff points (thresholds) that distinguish between continuous feeding and meal boundaries.

2.3 3. Exponential Decay Model

$$P(x + 1 \mid x) = a \cdot e^{-bx} \tag{3}$$

Purpose: Models the probability of adding an additional pellet ($P(x + 1 \mid x)$) based on the current cluster size (x). Here, a represents the initial probability, and b is the decay rate.

Role in Analysis: Quantifies how the likelihood of continuing pellet consumption decreases as the number of pellets in a cluster increases, providing a mathematical framework for understanding feeding behavior.

2.4 4. Log Transformation for Linear Regression

$$\ln P(x + 1 \mid x) = \ln a - bx \tag{4}$$

Purpose: Linearizes the exponential decay model to facilitate parameter estimation using linear regression techniques.

Role in Analysis: Enables the use of linear regression to estimate the parameters a and b by fitting a straight line to the transformed data.

2.5 5. Conditional Probability Calculation

$$P(x+1 \mid x) = \frac{\text{Number of } (x+1)\text{-pellet clusters}}{\text{Number of } x\text{-pellet clusters}} \tag{5}$$

Purpose: Determines the empirical probability of adding one more pellet given that a cluster of size x has been consumed.

Role in Analysis: Provides the observed probabilities that are used to fit the exponential decay model and to identify significant drops in pellet addition likelihood.

2.6 6. Threshold Determination Using Maximum Gap Method

$$\theta = P(x+1 \mid x) - \epsilon \tag{6}$$

Purpose: Sets a threshold (θ) just above the point of maximum drop in conditional probabilities to distinguish between typical meals and mega meals.

Role in Analysis: Facilitates the classification of pellet clusters into meals, snacks, and mega meals based on the identified threshold, ensuring a data-driven approach to meal boundary definition.

2.7 7. Confidence Interval Calculation (Wilson Score Interval)

$$\text{CI}_{\text{Wilson}} = \frac{1}{1+\frac{z^2}{n}} \left(\hat{p} + \frac{z^2}{2n} \pm z\sqrt{\frac{\hat{p}(1-\hat{p})}{n} + \frac{z^2}{4n^2}} \right) \tag{7}$$

Purpose: Estimates the 95% confidence intervals for the parameters a and b to assess their reliability and statistical significance.

Role in Analysis: Provides a measure of uncertainty around the estimated parameters, ensuring that the model's predictions are robust and dependable.

3 Summary of Key Findings Incorporated

- **Number of Mice Selected:** $n = 11$
- **Fitted Parameters:** $a = 142.74$, $b = 0.29$
- **Model Fit Quality:** $R^2 = 0.9760$ (Indicating an excellent fit)

4 Important Considerations

- **Parameter a Interpretation:** The parameter a in the exponential decay model typically represents a probability and should lie within the [0,1] range. However, in the provided findings, $a = 142.74$ exceeds this range, suggesting that the model may be scaled differently or that additional normalization is required. It's essential to ensure that parameters are interpreted correctly to maintain the validity of probability estimates.
- **High R^2 Value:** An R^2 of 0.9760 indicates that the exponential decay model explains 97.60% of the variability in the conditional probability $P(x+1 \mid x)$. This high value suggests that the model is highly effective in capturing the feeding behavior dynamics.

5 Conclusion

The formulas outlined above form the backbone of the mathematical and statistical framework used to analyze pellet consumption data in Order 1 mice during the NR phase. Each formula plays a critical role in ensuring that meal boundaries are defined accurately and objectively based on observed feeding patterns. Proper application and interpretation of these formulas are essential for deriving meaningful insights into the feeding behaviors of mice.